Identification and Estimation of Conditional Means from Aggregate Data

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January 26, 2025

I work on…

Questions in **social science**…	…which lead to **methodological** problems
• Quantifying **gerrymandering**	→ **sampling** from complex spaces
• Measuring **racial disparities**	→ post-**record linkage** inference
• Predicting the impacts of **electoral reform**	→ observational **causal inference**
• Understanding **public opinion** and forecasting **election results**	→ missing and coarsened data, including **ecological inference** (this talk)

software; papers & partnerships in substantive fields

Identification and Estimation of Conditional Means from Aggregate Data

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Joint work with **Shiro Kuriwaki** (Yale)

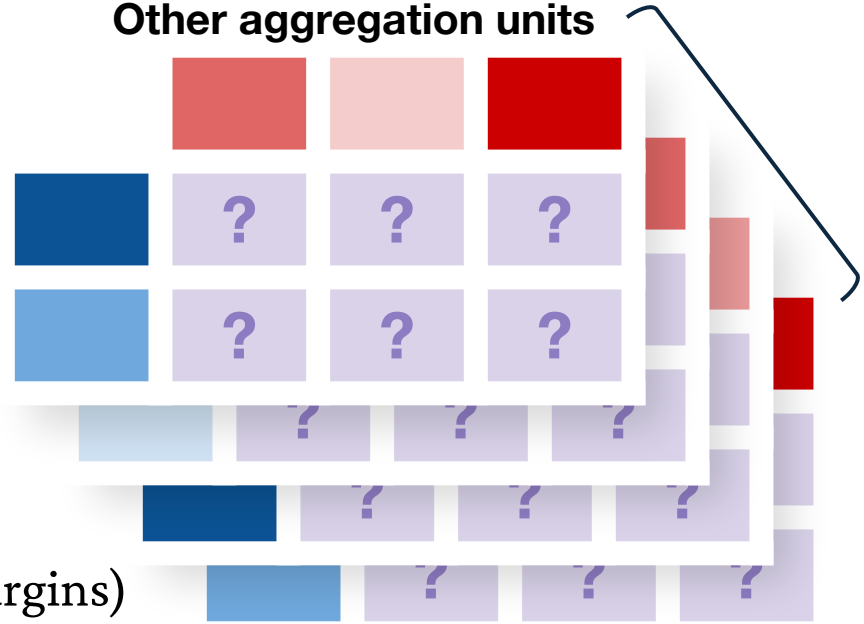
Motivation: Racially polarized voting

- Key question in **voting rights** cases: do voters of different **races** prefer different **candidates?**
- Current **Supreme Court** case over the role of race in drawing legislative districts
- **New factual test** for future voting rights cases: do voters of different races ***within the same party*** prefer different candidates?
- **Neither** question is statistically straightforward!

Callais v. Louisiana:

JUSTICE ALITO: I mean, that could be—that could be **easily analyzed by statistics** to see whether Black—whether white Democrats vote for **Black Democratic candidates** at a **lower rate** than they do for **white Democratic candidates**... It's easy to isolate race from that—from that to see whether there really is **racially polarized voting as opposed to partisan polarized voting.**

Ecological inference: The challenge of aggregate data



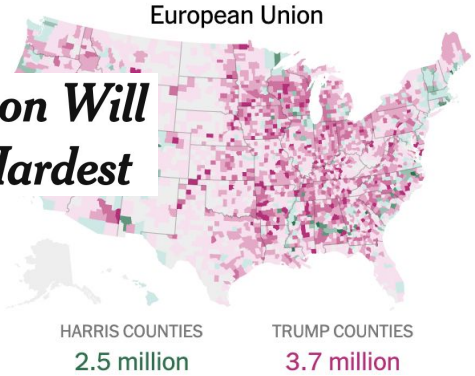
Aggregation unit
(voting precinct)

- **Goal:** learn $\beta_j = \mathbf{E}[B_j] = \mathbf{E}[Y \mid X_j = 1]$
- **Challenge:** only observe $\bar{Y}$ and $\bar{\mathbf{X}}$ (margins) due to **secret ballot**
- **Idea:** patterns in $\bar{Y}$ and $\bar{\mathbf{X}}$ are informative

Ecological inferences are unavoidable

- Election returns: **secret ballots** tabulated at precinct level
- **Epidemiological** studies: exposure, disease by geography
- Some individual-level linkages are **banned by law**, e.g. U.S. Treasury and U.S. Census data
- **Historical** data where only published aggregates survive

Trade War Retaliation Will Hit Trump Voters Hardest



See Which Neighborhoods and Groups Drove Mamdani's Success in the N.Y.C. Primary

PRECINCTS WITH...	ANDREW CUOMO	ZOHRAN MAMDANI	BRAD LANDER
Higher income residents 24% of precincts	30%	42%	20%
Middle income residents 49% of precincts	37%	47%	7%
Lower income residents 24% of precincts	49%	38%	3%

Current EI methods are limited

- **Goodman** (1953, 1959) **regression:** linear regression of $\bar{Y}$ on $\bar{\mathbf{X}}$
  → based on **strong** + **implausible** assumption
  → often yields impossible estimates

- **Partial identification** (e.g., Cross & Manski 2002) for bounded outcomes
  → bounds are often too wide to be useful

- **King's EI** (1997): hierarchical model for missing data, incorporating bounds
  → based on **same assumptions** as Goodman regression
  → relies on parametric assumptions; **no** inference guarantees
  (Freedman et al. 1998)
  → **most used in practice**, including extensions to >2 predictors
  (Rosen, Jiang, King, & Tanner 2001)

This talk

When can we learn from aggregate data?

How can we **estimate** regressions in aggregate data?

What happens when our **assumptions** are violated?

Does it **matter**? Does it **work**?

Application to racially polarized voting in Louisiana

→ First **nonparametric identification** result

→ New sieve **Debiased ML estimator**

→ First **sensitivity analysis** for ecological inference

→ **Yes**: simulation study and **validation data**

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Setup

- Individual outcome Y, predictor $\mathbf{X}$, "geography" G
- Aggregate outcome $\bar{Y}_g = \mathbf{E}_n[Y \mid G = g]$ — vote share
- Aggregate predictor $\bar{\mathbf{X}}_g = \mathbf{E}_n[\mathbf{X} \mid G = g]$ — % in each racial group
- Covariates $\mathbf{Z}_g$ — e.g., income or age distribution
- Observation count N_g — number of voters
- Local conditional mean $\mathbf{B}_g$ — vote share by racial group (j)
  $B_{gj} = \mathbf{E}_n[Y \mid X_j = 1, G = g]$
- Estimand $\beta_j = \mathbf{E}[\bar{X}_{Gj}\, \mathbf{B}_{Gj}] \,/\, \mathbf{E}[\bar{X}_{Gj}] = \mathbf{E}[Y \mid X_j = 1]$
  (geographies weighted by N_g)

The key accounting identity

- In each **geography,**

$$\textbf{\% Dem} = (\textbf{\% Dem} \text{ among White voters})(\textbf{\% white}) + (\textbf{\% Dem} \text{ among Black voters})(\textbf{\% black})$$

$$\bar{Y} = B_1 \bar{X}_1 + B_0 \bar{X}_0 \qquad \Rightarrow \qquad \bar{Y} = \mathbf{B}^{\mathsf{T}} \bar{\mathbf{X}}$$

- *cf.* causal inference

$$Y = Y(1)\,T + Y(0)\,(1 - T)$$

- Holds **exactly** without error $\longrightarrow$ leads to **bounds** when Y is a proportion
- **Problem:** each observation contributes multiple unknowns

Identifying assumptions

Assumption CAR
(Coarsening at random)

$$\mathbf{E}[\mathbf{B} \mid \mathbf{Z}, \bar{\mathbf{X}}, N] = \mathbf{E}[\mathbf{B} \mid \mathbf{Z}]$$

i.e.,

$$\mathbf{E}[\mathbf{B} \mid \mathbf{Z}, \bar{\mathbf{X}}] = \mathbf{E}[\mathbf{B} \mid \mathbf{Z}] \text{ and}$$
$$\mathbf{E}[\mathbf{B} \mid \mathbf{Z}, \bar{\mathbf{X}}, N] = \mathbf{E}[\mathbf{B} \mid \mathbf{Z}, \bar{\mathbf{X}}] \quad (opt.)$$

Theorem
(Nonparametric identification)

Under CAR, β is identified as

$$\beta_j = \mathbf{E}[\underbrace{u(N, \bar{\mathbf{X}})}_{\substack{\text{weighting term} \\ = N\bar{\mathbf{X}}/E[N\bar{\mathbf{X}}]}} \underbrace{\mathbf{E}[\bar{Y} \mid \bar{X}_j = 1, \mathbf{Z}]}_{\substack{\text{regression evaluated at} \\ \bar{X}_j = 1 \text{ and averaged} \\ \text{over covariates}}}]$$

e.g., local White preference for Democrats *indep. of* racial composition of area, *given* covariates

An individual perspective

- CAR is stated in terms of **aggregate variables** and thus may be **hard to evaluate** or interpret
- Alternative **individual-level** assumption:

  Assumption CAR-IND. $\mathbf{E}[Y \mid G, \boldsymbol{X}, \boldsymbol{Z}_G] = \mathbf{E}[Y \mid \boldsymbol{X}, \boldsymbol{Z}_G]$

  e.g., voting *indep. of* residence *given* race, income, age, etc.
- Stronger than CAR, so sufficient for identification.

  Theorem. CAR-IND $\Rightarrow$ CAR.

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First steps: partial linearity

- Identification result: Estimand can be written as

$$\beta_j = \mathbf{E}[u(N, \bar{\mathbf{X}})\, \gamma([0, 1, \ldots, 0], \mathbf{Z})]$$

  where $\gamma(\bar{X}, Z) := E[\bar{Y} \mid \bar{X}, Z]$ is the CEF of $\bar{Y}$

- Accounting identity (LTE) + CAR imply

$$\bar{Y} = \mathbf{B}^\top \bar{\mathbf{X}} = (\eta(\mathbf{Z}) + \varepsilon)^\top \bar{\mathbf{X}} = \eta(\mathbf{Z})^\top \bar{\mathbf{X}} + \varepsilon^\top \bar{\mathbf{X}},$$

  where $\eta(\mathbf{Z}) := \mathbf{E}[\mathbf{B} \mid \mathbf{Z}]$ and ε is mean-0 and exogenous

- When $\mathbf{Z}$ is null, $\eta(\mathbf{Z}) = \boldsymbol{\beta}$ is constant, and we recover Goodman

- **Key observation:** CEF of $\bar{Y}$ has $\bar{\mathbf{X}}$-linear structure

A functional perspective

- CEF $\gamma(\bar{\mathbf{X}}, \mathbf{Z})$ is of the form $\eta_0(\mathbf{Z})^\intercal \bar{\mathbf{X}}$ for $\eta_0 \in L^2(\mathbf{Z})^{\otimes|\mathscr{X}|}$
  - Functions like this are a linear **subspace** Γ of $L^2(\bar{\mathbf{X}}, \mathbf{Z})$
- **Theorem.** Γ is **closed**, and the functional $\gamma \longmapsto \mathbf{E}[l(N, \bar{\mathbf{X}})\, \gamma([0, 1, \ldots, 0], \mathbf{Z})]$ is **mean-square continuous** in γ …

  … under a **positivity** assumption that ensures sufficient variation in $\bar{\mathbf{X}}$:
- **Assumption POS**. There exists a $\delta > 0$ such that $f(\bar{\boldsymbol{x}}, \boldsymbol{z}) > \delta f(\boldsymbol{z})$.
  - Intuition: if distribution $\bar{\mathbf{X}} \mid \mathbf{Z}$ isn't supported everywhere, can't extrapolate
- Corollary: **Riesz representation** theorem guarantees RR (weights) $\alpha_j(\bar{\mathbf{X}}, \mathbf{Z})$ in **same** partially linear subspace which satisfy $\beta_j = \mathbf{E}[\alpha_j \gamma] = \mathbf{E}[\alpha_j \bar{Y}]$
  - Analog of IPW representation of ATE in causal: α is inverse prop. score

Debiased ML estimation for EI

- Which representation? $\beta_j = \mathbf{E}[u(N, \bar{\mathbf{X}})\, \gamma([0, 1, \ldots, 0], \mathbf{Z})] = \mathbf{E}[\alpha_j(\bar{\mathbf{X}}, \mathbf{Z})\, \bar{Y}]$
- Both! Fit $\hat{\gamma}$ and $\hat{\alpha}$ **within** the restricted function class Γ
- Estimate β via Neyman-orthogonal score:

$$\hat{\beta}_j = \mathbf{E}_n[\underbrace{u(N, \bar{\mathbf{X}})\, \hat{\gamma}([0, 1, \ldots, 0], \mathbf{Z})}_{\text{Plug-in estimate}} + \underbrace{\hat{\alpha}_j\, (\bar{Y} - \hat{\gamma}(\bar{\mathbf{X}}, \mathbf{Z}))}_{\text{RR-weighted residual}}]$$

- **No** sample splitting: **algorithmic stability** suffices (Chen et al. 2022)
- **Need**: well-behaved $\mathbf{Z}$, $|\hat{\gamma} - \gamma|_2 = o_{\mathrm{P}}(n^{-1/4})$ and $|\hat{\alpha}_j - \alpha_j|_2 = o_{\mathrm{P}}(n^{-1/4})$
- *Then*: consistency, asymptotic normality, semiparametric efficiency

Nuisance function estimation

- To estimate $\hat{\gamma}$ and $\hat{\alpha}_j$ within restricted function class Γ, **interact** predictors $\bar{\mathbf{X}}$ with **sieve basis** expansion $\Phi(\mathbf{Z})$
- Fit regression (in practice, ridge) of $\bar{Y}$ on $\bar{\mathbf{X}} \otimes \Phi(\mathbf{Z})$
  - Different loss function for estimating $\alpha_j \Rightarrow$ **closed form** available
  - **Bounded** outcomes $\Rightarrow$ **quadratic program** to fit regression γ
- We show standard sieve **regularity conditions** yield standard **estimation rates** within Γ (with or without penalization)

Theorem 6. *Under BND, POS, and SR, $\hat{\gamma}_m$ and $\hat{\alpha}_{mj}$ exist uniquely with probability approaching one as $m \to \infty$, and for all $j \in \mathcal{X}$, we have for the unpenalized estimators that*

$$\|\hat{\gamma}_m(0) - \gamma_0\| = O_{\mathbb{P}}\left(\rho_m + \sqrt{J_m/m}\right) \quad \text{and} \quad \|\hat{\alpha}_{mj}(0) - \alpha_{0j}\| = O_{\mathbb{P}}\left(\rho_m + \sqrt{J_m/m}\right).$$

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Sensitivity analysis for CAR

- **So far,** estimates have required **CAR** to hold
- What happens when we are **missing** a key **confounder** A ?
  Compare estimate with and without A controlled for
- **Sensitivity analysis** based on DML (Chernozhukov et al. 2024)

Theorem 10. *When CAR holds conditional on A_G, then*

$$\left|\beta_j^* - \beta_j\right| \leq \rho S C_\gamma C_\alpha,$$

where $\rho := \left|\mathrm{Cor}\left(\gamma_0^A - \gamma_0, \alpha_{0j}^A - \alpha_{0j}\right)\right|, \qquad S^2 := \mathbb{E}\left[\left(\overline{Y} - \gamma_0\left(\overline{X}, Z\right)\right)^2\right]\mathbb{E}\left[\alpha_{0j}\left(\overline{X}, Z\right)^2\right],$

$$C_\gamma^2 := \frac{\mathbb{E}\left[\left(\gamma_0^A - \gamma_0\right)^2\right]}{\mathbb{E}\left[\left(\alpha_{0j}^A - \alpha_{0j}\right)^2\right]} = R^2_{\overline{Y} \sim A | \overline{X}, Z}, \quad \text{and} \quad C_\alpha^2 := \frac{\mathbb{E}\left[\alpha_{0j}^{A^2}\right] - \mathbb{E}\left[\alpha_{0j}^2\right]}{\mathbb{E}\left[\alpha_{0j}^2\right]} = \frac{1 - R^2_{\alpha_{0j}^A \sim \alpha_{0j}}}{R^2_{\alpha_{0j}^A \sim \alpha_{0j}}}.$$

Sensitivity analysis

Theorem 10. *When CAR holds conditional on A_G, then*

$$\left|\beta_j^* - \beta_j\right| \leq \rho S C_\gamma C_\alpha,$$

where $\rho := \left|\mathrm{Cor}\left(\gamma_0^A - \gamma_0, \alpha_{0j}^A - \alpha_{0j}\right)\right|, \qquad S^2 := \mathbb{E}\left[\left(\overline{Y} - \gamma_0\left(\overline{X}, Z\right)\right)^2\right]\mathbb{E}\left[\alpha_{0j}\left(\overline{X}, Z\right)^2\right],$

$$C_\gamma^2 := \frac{\mathbb{E}\left[\left(\gamma_0^A - \gamma_0\right)^2\right]}{\mathbb{E}\left[\left(\alpha_{0j}^A - \alpha_{0j}\right)^2\right]} = R^2_{\overline{Y} \sim A | \overline{X}, Z}, \quad \text{and} \quad C_\alpha^2 := \frac{\mathbb{E}\left[\alpha_{0j}^{A^2}\right] - \mathbb{E}\left[\alpha_{0j}^2\right]}{\mathbb{E}\left[\alpha_{0j}^2\right]} = \frac{1 - R^2_{\alpha_{0j}^A \sim \alpha_{0j}}}{R^2_{\alpha_{0j}^A \sim \alpha_{0j}}}.$$

- C_γ = how much residual variance in $\bar{Y}$ is explained by A
  C_α ≈ how much residual variance in $\bar{\mathbf{X}}$ is explained by A (very rough)
- Interpretation: **benchmark** parameters based on observed covariates
- **Display bias** bounds on **contour plot**

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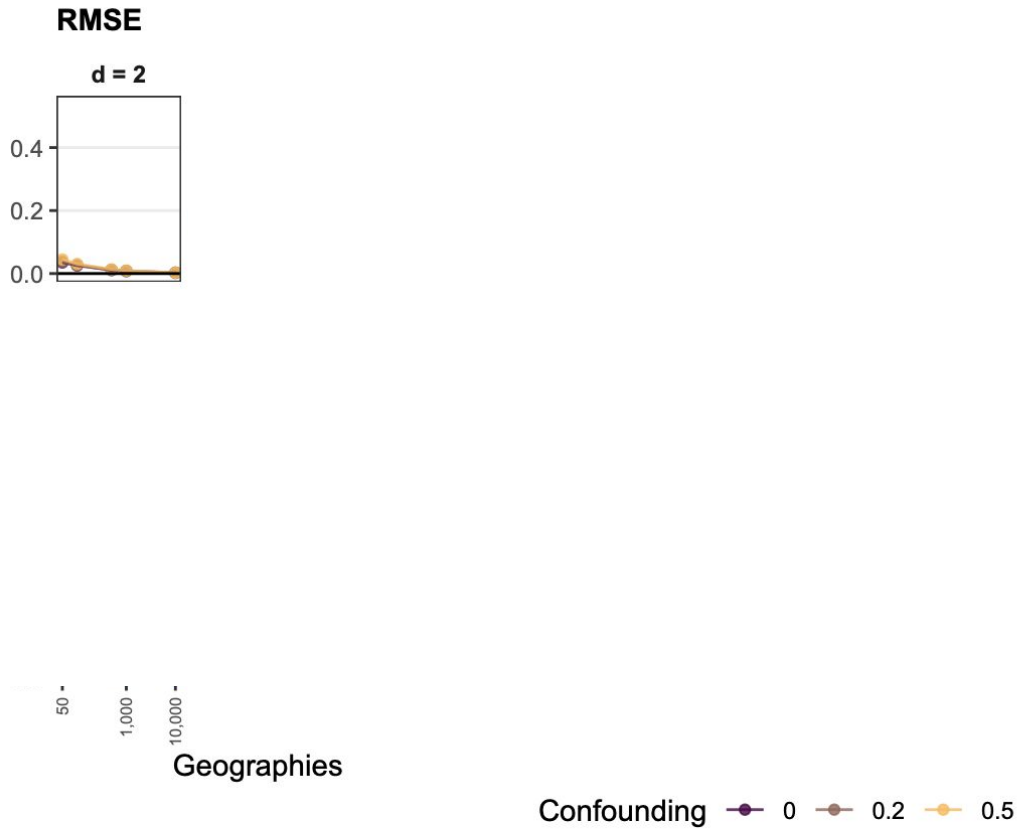
New method outperforms old methods

	Covariates?	RMSE	Coverage (50%)	Coverage (95%)	Time (s)
Proposed method	+				
Goodman (linear regression)					
King (1997; `ei`)	+				
King (1997; `ei`)					
RJKT (2002; `eiPack`)	+				
RJKT (2002; `eiPack`)					

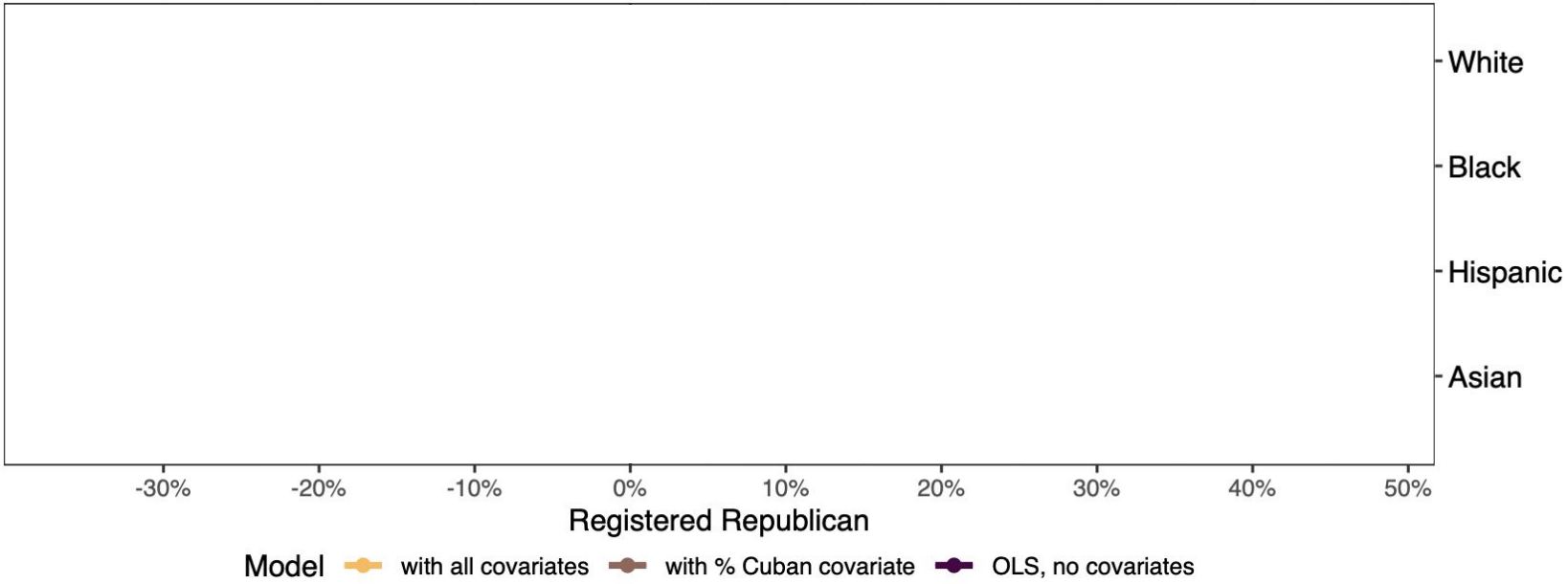
Table 1: **Comparison of methods on simulated data**. RMSE, coverage of 50% and 95% nominal confidence intervals, and average computation time (in seconds) for different methods on simulated data. A '+' in the covariates column indicates that the method controlled for confounding covariates. RJKT refers to Rosen et al. (2001).

New method is accurate & calibrated



Covariates matter!

- Miami-area voter registration data: **Y** = GOP registration, **X** = race
- **Ground truth** is observed



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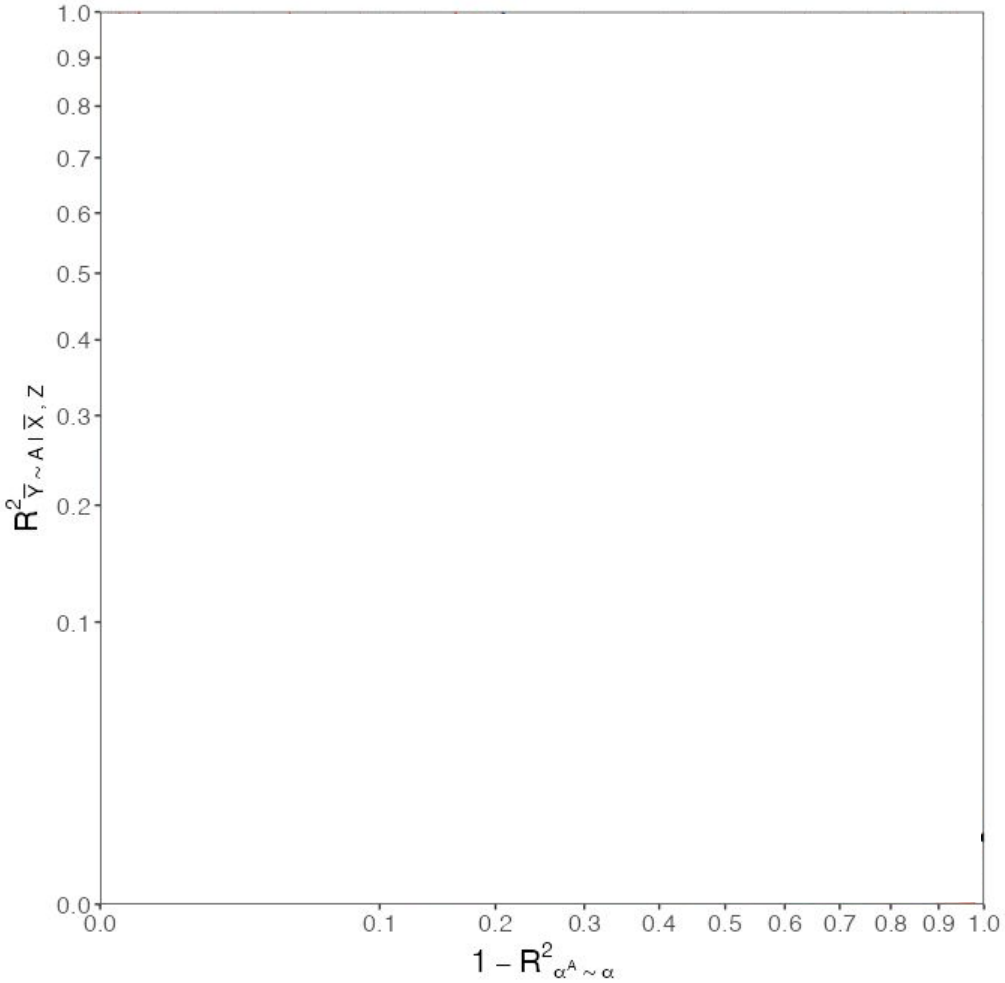
2022 Louisiana gubernatorial election

- Did **White** voters (58% of VAP) prefer **Chambers** (17.8% overall) to **Mixon** (13.2% overall) at a different rate than **Black** voters (30% of VAP)?
- $\mathbf{E}[\mathbf{1}\{\text{CHAMBERS}\} - \mathbf{1}\{\text{MIXON}\} \mid \text{BLACK}] - \mathbf{E}[\mathbf{1}\{\text{CHAMBERS}\} - \mathbf{1}\{\text{MIXON}\} \mid \text{WHITE}]$

Results

- **Covariates:** population, density, turnout, partisanship, geography
- **Racially polarized voting**: Democratic support is
  - Among White voters: **–2.25pp** (0.180)
  - Among Black voters: **+9.42pp** (0.393)
- **Sensitivity analysis** with benchmarked covariates shows robustness



Summary

TRY IT OUT:

`corymccartan.com/seine`

Paper

- **Existing** EI methods ≈ **causal inference** by comparing **means**
- Controlling for **covariates** allows for **credible EI**
- New, modern, estimation methods are more **accurate** and **efficient**
- **Sensitivity analysis** recommended to probe key assumptions

Local estimates

- Accounting identity: **no error** term
- **Project** regression residuals onto accounting identity **constraint**
- Can build (asymptotically) **valid confidence intervals**

$$R_g'^{\alpha} := \left\{ b \in H'\left(\overline{X}_g, \overline{Y}_g\right) : \left(b - \hat{B}'_g\right)^\top \left(\hat{\Pi}_g \hat{\Sigma}(Z_g) \hat{\Pi}_g\right)^+ \left(b - \hat{B}'_g\right) \leq \frac{d-1}{\alpha} \right\}$$

Theorem 9. *Suppose that* $\hat{\eta} \xrightarrow{p} \eta_0$ *and* $\hat{\kappa} \xrightarrow{p} \kappa_0$ *pointwise. Then for* $0 < \alpha < 1$, *as* $m \to \infty$,

$$\mathbb{P}\left(B_g \in R'^{\alpha}_g\right) \geq 1 - \alpha + o(1).$$